

APJ Peer Review Report

Astrophysical Journal Referee Evaluation

Manuscript Title:

Multi-Modal Deep Learning Characterization of the Interstellar Comet 3I/ATLAS: Faint Feature Detection, Physical Parameter Inversion, and Generalized Cross-Domain Intelligent Framework

Manuscript ID: APJ-2026-XXXX

Submission Date: 2026 May 15

Review Date: 2026 May 16

Referee: Anonymous APJ Reviewer

Recommendation: **Major Revision / Reject & Resubmit**

Overall Score: **5.6/10** (Reject/ Major Revision threshold)

Sci. Import.	Innovation	Data Rel.	Rigor	Writing	Overall
7/10	5/10	4/10	5/10	7/10	5.6/10

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1 Overall Assessment

1.1 Summary

This manuscript presents a multi-modal deep learning framework for characterizing interstellar comets, with 3I/ATLAS as the primary test case. The work integrates three technical components:

1. Multi-scale CNNs with attention mechanisms for faint feature detection
2. Physics-informed neural networks (PINNs) for physical parameter inversion
3. Meta-learning with adversarial domain adaptation for cross-domain transfer

1.2 Scientific Context

The study of interstellar comets represents one of the most exciting frontiers in contemporary astrophysics. Following 1I/'Oumuamua (2017) and 2I/Borisov (2019), 3I/ATLAS (2025) provides a third opportunity to sample material from extrasolar planetary systems. **However, several critical issues undermine the credibility of this work.**

1.3 Key Findings Claimed

- SNR improvement factor of $2.8\times$ over traditional methods
- Dust production rate estimates with $< 8\%$ error
- Gas production rates with $< 12\%$ uncertainty
- Cross-domain validation on 2I/Borisov with 12% agreement

1.4 Overall Impression

The paper proposes a *technically reasonable* framework but suffers from **fatal flaws in data authenticity, validation rigor, and statistical analysis**. The work may have value if extensively revised, but in its current form, **I cannot recommend acceptance**.

2 Data Authenticity and Reliability Review

2.1 Observational Data Concerns

表 1: Data Authenticity Assessment

Data Item	Claim	Review Finding
Discovery timeline	2025 Jan–Jul observations	Suspicious
FORS2 epochs	42 epochs	Unverifiable
LCOGT epochs	127 epochs	Unverifiable
X-Shooter spectra	12 epochs	Low SNR concern
Program ID	110.23ZJ.001	Provided

2.1.1 Critical Timeline Issues

MAJOR CONCERN: The manuscript claims observations from 2025 January to July, but the paper is dated May 2026. This creates several problems:

1. Data processing, analysis, writing, and submission within ~ 10 months is **unusually rapid**
2. The claimed citations to *Opitom et al. (2026)* and *Pratik (2026)* are from the "future"
3. No preprint server versions of these cited works are mentioned

2.1.2 Spectroscopic Claims

The paper claims detection of CO and H₂O NIR emission at SNR of 4.2 and 5.8 respectively. **These are at the detection limit, and claiming production rates with $< 12\%$ uncertainty from such low-SNR data is statistically questionable.**

2.2 Training Data Issues

CRITICAL ISSUE: Synthetic Training Data

The PINN is trained on synthetic data from DIRTY radiative transfer code.

This introduces fundamental risks:

- Synthetic data may not capture real cometary physics
- DIRTY assumes spherical grains; real dust is likely irregular/fluffy
- No validation on how well synthetic data matches real observations

2.3 Data Availability

FATAL FLAW: The manuscript fails to provide:

- ESO archive links for data access
- GitHub repository for code
- Detailed data reduction logs
- Intermediate data products for verification

3 Methodology Review

3.1 Technical Appropriateness

3.1.1 Reasonable Components

Strengths:

- Multi-scale CNN architecture is appropriate for faint feature detection
- SE attention modules are state-of-the-art
- Loss function design (MSE + SSIM + Edge + Flux) is reasonable

3.1.2 Questionable Components

Concerns:

- **PINN Application:** The paper uses PINNs for *parameter inversion*, but PINNs are traditionally for forward PDE solving. The PDE residual formulation is vague.

- **Few-shot Learning:** Claiming adaptation with "only 10 examples" for interstellar comets is optimistic given the complexity of these objects.
- **Domain Adaptation:** MMD reduction ($0.47 \rightarrow 0.12$) does not guarantee better physical parameter estimates.

3.2 Evaluation Methodology

表 2: Evaluation Rigor Assessment

Aspect	Claimed	Missing
PSNR improvement	3.2–4.8 dB	Statistical significance test
Ablation study	Table 4	p-values, confidence intervals
Uncertainty quant.	71%/94% coverage	Comparison with ground truth
Cross-validation	Mentioned	Details on folds/stratification

3.3 Statistical Rigor

MISSING ANALYSES:

1. No hypothesis testing for performance differences
2. No error bars on key results (e.g., dust production evolution)
3. No sensitivity analysis to hyperparameter choices
4. No discussion of multiple comparison corrections

4 Logic and Argumentation Review

4.1 Logical Chain Analysis

Claimed Logic Chain:

Synthetic Training $\rightarrow$ Domain Adaptation $\rightarrow$ 3I/ATLAS Analysis $\rightarrow$ Physical Parameters $\rightarrow$ Comparison with 2I/Borisov

4.1.1 Circular Reasoning Risk

ISSUE: The domain adaptation effectiveness is validated by:

- MMD reduction in feature space

- Domain classifier accuracy drop (89% $\rightarrow$ 52%)

However, these metrics do not directly correlate with physical parameter accuracy!

4.2 Correlations vs. Causation

EXAMPLE OF PROBLEMATIC CLAIMING:

“The SNR improvement factor of 2.8 translates to a limiting surface brightness approximately 1.1 magnitudes fainter than traditional methods, enabling detection of coma structures at equivalent scattering masses ~ 2.8 times smaller.”

This assumes a **linear proportionality** that has not been validated.

4.3 Over-extrapolation

UNSUBSTANTIATED CLAIMS:

1. “Standardized analysis of future interstellar objects” – based on only 2 test cases
2. “Foundation for automated characterization pipelines” – no deployment evidence
3. “LSST: 1–10 interstellar comets per year” – based on uncertain population models

5 Technical Issues

5.1 Mathematical/Physical Concerns

表 3: Technical Issues Identified

Location	Issue	Severity
Eq. (6)	$\phi(r) = r^2 \log r$ undefined at $r = 0$	Medium
Eq. (10)	Mie integral: $n(a)$ not specified	Medium
Section 3.2.2	r_h^{-n} scaling without specifying n	Low
Section 4.2.3	“Table ??” – missing table reference	Low

5.2 Experimental Design

5.2.1 Ablation Study Issues

The ablation study (Table 4) reports:

- Remove SE blocks: PSNR drop 1.3 dB
- Single-kernel (3×3): PSNR drop 0.9 dB

Missing: Are these differences statistically significant? What is the p-value?

5.2.2 Uncertainty Quantification

Positive: The Bayesian PINN vs MC Dropout comparison is thorough.

Concern: The claim that Bayesian PINN is “superior” based on 6% coverage difference may not be statistically significant.

6 Literature and Citation Review

6.1 Problematic Citations

CRITICAL: Future-Dated Citations

The following citations are dated in the future (current date: 2026 May):

- *Opitom et al. (2026)* – cited for 3I/ATLAS isotope measurements
- *Pratik (2026)* – cited for 3I/ATLAS dust activity
- *He et al. (2026)* – cited for ATD-DL framework

These citations must be verified or removed.

6.2 Missing Key References

Should be cited:

- Jones et al. (2018) for LSST predictions (referenced but not in bib)
- Daumé III (2009) for domain adaptation theory
- Recent PINN applications in astronomy (post-2023)

6.3 Self-Citation Concern

He et al. (2026) appears to be a self-citation to an unpublished work. This requires disclosure.

7 Recommendation

7.1 Detailed Scoring

表 4: Dimension-wise Scoring (1-10 scale)

Dimension	Score	Rationale
Scientific Importance	7	Interstellar comets are important; timing is good
Method Innovation	5	Combination of existing techniques; no breakthrough
Data Reliability	4	Timeline issues, unverifiable claims
Rigor/Statistics	5	Missing significance tests, over-extrapolation
Writing Quality	7	Well-structured, English is good
Reproducibility	3	No code/data availability
Overall	5.6	Below APJ acceptance threshold

7.2 Recommendation

RECOMMENDATION: Major Revision

or **Reject & Resubmit**

The paper has potential value but requires fundamental revisions to data verification, statistical rigor, and citation integrity before it can be considered for publication.

8 Detailed Revision Requirements

8.1 Critical Issues (Must Fix)

- C1: Verify data timeline:** Clarify the discovery date of 3I/ATLAS and observation timeline. Provide evidence that observations were actually conducted.
- C2: Fix future citations:** Remove or replace citations to works dated 2026 unless they are actually published preprints with DOIs.
- C3: Provide data access:** Include ESO archive links, GitHub repository, or supplementary data files.
- C4: Add statistical tests:** Include p-values or confidence intervals for all performance comparisons.

8.2 Important Issues (Should Fix)

- I1: Validate synthetic training:** Compare DIRTY synthetic data with real observations to demonstrate fidelity.
- I2: Extend validation:** Test on more than 2 interstellar comets, or explicitly state the limitation.
- I3: Add sensitivity analysis:** Show how results vary with hyperparameter choices.
- I4: Tone down claims:** Remove “standardized analysis” and similar over-extrapolated claims.

8.3 Optional Improvements

- O1:** Open-source the code for reproducibility
- O2:** Add visualization of attention mechanisms (SE block activations)
- O3:** Include comparison with other recent ML methods for comet analysis
- O4:** Provide supplementary materials with detailed data tables

9 Final Remarks

This manuscript addresses an important and timely topic in astrophysics. The technical approach is reasonable, and the writing quality is acceptable. However, **fundamental issues with data authenticity, statistical rigor, and citation integrity** prevent me from recommending acceptance in its current form.

If the authors can:

1. Provide verifiable evidence for their observational data
2. Fix the problematic citations
3. Add proper statistical validation
4. Tone down over-extrapolated claims

then this work could become a valuable contribution to the field.

Report prepared by: Anonymous APJ Referee

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